

Switching Loss Optimization and Soft Switching Range Extension of a Current-fed Triple Active Bridge Converter

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Abstract—On-board charger (OBC) and auxiliary power module (APM) are crucial power electronic components in electric vehicles. The OBC converts AC input into DC output to charge the High-Voltage (HV) propulsion battery; while the APM connects the HV side and Low-Voltage (LV) side to charge the LV battery. To reduce costs, minimize size, and enhance power density, this paper studied a three-port current-fed triple-active bridge (CF-TAB) converter for integrating the OBC and the APM. This converter offers a streamlined structure and simplifies transformer design compared to traditional integration methods. In this paper, a five-control-variable optimal control scheme utilizing a look-up table method is proposed for the converter with the objectives of minimizing the switching loss of the converter. In addition, a hardware experimental platform for 3kW/200V-500V OBC plus 1kW/8V-16V APM is built and allows for the verification of the proposed control scheme's effectiveness. The results indicate that, in comparison to conventional DPS control, the new control scheme significantly reduces switching losses and extends the soft-switching range.

Keywords—Current-fed triple active bridge (CF-TAB); Switching loss optimization; Soft switching; Integration of OBC and APM for electric vehicles.

I. INTRODUCTION

With the continuous enhancement of awareness of energy conservation and environmental protection, electric vehicles have become an important solution to the problems of energy shortage and environmental pollution due to their low emission and high efficiency advantages compared with traditional vehicles. The OBC and APM are crucial power electronic components in electric vehicles. The OBC serves as the interface between the power grid and the high-voltage battery, converting AC input into DC output for charging the high-voltage battery. Meanwhile, the APM connects the high-voltage system to the low-voltage system within the vehicle, stepping down the voltage from the high-voltage DC bus to the low-voltage bus to charge the low-voltage battery. The

integration of OBC and APM can provide further cost savings and size reduction. In order to realize the integration of OBC and APM, so many scholars have conducted lots of research.

A three-port current-fed triple active bridge (CF-TAB) converter is proposed in [1], which combines OBC and APM in both electrical and magnetic aspects. However, only open-loop power transmission characteristics were verified, and the control strategy and optimization are not studied in depth. An isolated three-port bidirectional DC-DC converter is proposed in [2]. Phase shift plus duty cycle control is discussed in this paper to optimize the system behavior. The control scheme aims to ensure the minimum overall loss of the system. Reference [3] aims to derive and analyze the ZVS conditions for TAB converter. An optimized five-variable-based control strategy is proposed in the literature to achieve minimum switching loss over a wide range of voltage gains and loads. However, the topology suffers the larger output current ripples and the higher turn ratio of the transformer. Reference [4] aims to improve the efficiency of TAB converter by minimizing conduction losses. The optimization is based on the minimum rms current, expressed in the generalized harmonic approximation technique, neglecting the soft switching of the switch transistors. A strategy to minimize conduction losses by utilizing the five control variables of the converter is proposed in [5], aiming to improve the efficiency by reducing the rms current of the low-voltage bridge. However, the loss of soft switching in part of the switch transistors is not taken into account. For the current-fed dual active bridge bidirectional DC-DC converter, an extended duty cycle control approach is introduced in [6]. Additionally, a voltage mismatch control with adaptive clamping voltage regulation is proposed, enabling ZVS for all switches across the entire load range.

A new modulation scheme is proposed in [7] to improve the efficiency over a wide range of voltage and load. The relationship between the duty cycles and the phase shifts of the quadratic full bridge is studied to reduce the power loss under

wide load variation. On the basis of the analysis of the working mode, a control loop using PWM and phase shift modulation scheme was designed. The switching losses of the previous and proposed modulation schemes are compared. The ZVS characteristics and current stresses of the CF-DAB converter are analyzed in [8]. On this basis, an algorithm is proposed to determine the given working conditions. For the CF-DAB converter, all possible switching modes are summarized from the combined perspective of the pulse-width modulation duty cycle and phase shift angle on both sides of the converter[9]. In this paper, a control strategy is proposed to make it work in the optimized mode. The research status of electric vehicle charging technology is expounded, and some predictions are made for possible future research directions[10].

An integrated power system for electric vehicles is proposed in [11], which integrates the OBC and the LDC. This paper provides a thorough theoretical analysis of the system's topology, operating modes, control strategies, and transformer design. A three-port bidirectional DC converter with staggered parallel boost ports tailored for electric vehicle distribution networks was proposed in [12]. This converter boasts low current ripple in its boost port, voltage-matched zero-voltage switching (ZVS), and a low transformer turn ratio. Lastly, [13] proposed a converter based on a reconfigurable current-fed dual active bridge (CF-DAB) to address the challenges posed by a wider input voltage range in the APM.

For the current-fed triple active bridge converter used in the integration of OBC and APM, a control strategy taking the extension of soft switching range and reducing the optimization objective function related to the switching loss into account in the same time is not found. Therefore, this paper presents an optimization method for the goal of expanding the soft switching range and reducing the optimization objective function related to the switching loss.

The topology description, power flow characteristic, related current derivation and proposed control strategy are analyzed in section II. In Section III, a hardware experimental platform is built to verify the effectiveness of the proposed control strategy for CF-TAB converter. Finally, concluding remarks are placed in section IV.

II. THEORETICAL ANALYSIS AND CONTROL STRATEGY

A. The Topology Description and Power Flow

1) Topology Description

The topology of the CF-TAB converter is shown in Fig. 1, which consists of three full-bridge units connected by a three-winding transformer T_r , whose turns ratio is $N_1:N_2:N_3$. S_1 - S_4 , S_5 - S_8 and S_9 - S_{12} are the MOSFETs of the three full-bridge units respectively. L_1 , L_2 and L_3 are the power transmission inductors of the converter. L_{o1} and L_{o2} are the output inductors of the HV side, L_{o3} and L_{o4} are the output inductors of the LV side, $L_{o1} = L_{o2}$, $L_{o3} = L_{o4}$.

Five control variables are involved in the operation of the proposed converter, i.e. D_p , D_H , D_L , ϕ_{12} , and ϕ_{13} . D_p , D_H and D_L are the duty cycles of the switch transistors, S_1/S_3 , S_5/S_7 and S_9/S_{11} , respectively, and the duty cycles of the upper and lower switch transistors of the same bridge arm are kept

complementary at the same time. ϕ_{12} and ϕ_{13} are the phase shifts for the HV and LV side, referred to the primary side. Fig. 2 shows the PWM signals of all switch transistors within single switching period and the waveforms of the voltages, u_p , u_H , and u_L , corresponding to three ports of the transformer, containing all five control variables. The ranges of the duty cycles (D_p , D_H , D_L) are 0~0.5, while the phase shifts (ϕ_{12} , ϕ_{13}) are 0~0.5 π , when the power flows from the primary side to HV and LV side.

For the secondary side, taking the HV side as an example, the switch transistors S_5 - S_8 not only makes up a dual-active bridge circuit with the primary side to assume the function of power transmission, but also forms a interleaved parallel Buck converter with the output inductors L_{o1} and L_{o2} to realize the conversion from the clamping capacitor C_{HV} voltage to the output voltage.

The same is true for the secondary LV side.

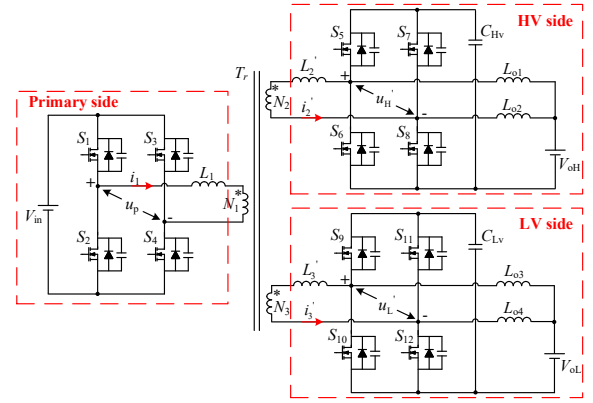


Fig. 1. Topology of the CF-TAB converter.

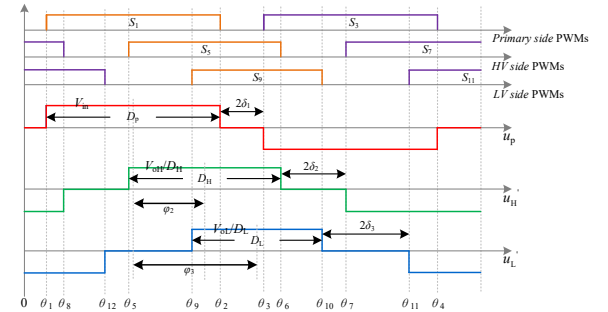


Fig. 2. PWM signals and transformer port voltage waveforms.

This topology reduces the turn ratio and facilitates the design of the transformer through the multiplexing of the switch transistors. In addition, the output current ripple is significantly suppressed due to the interleaving of output inductors.

2) Equivalent Circuit of a Three-Winding Transformer

The equivalent circuit of a three-windings transformer and its Y/ Δ type equivalent circuit referred to primary side are shown in Fig. 3.

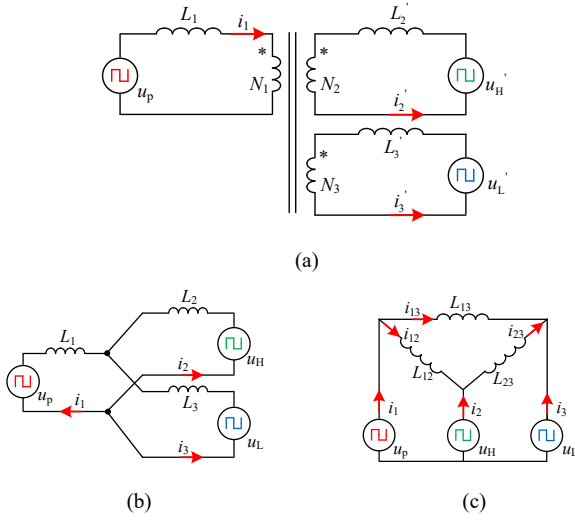


Fig. 3. Transformer equivalent circuits and Y/Δ transformations (a) The transformer equivalent circuit (b) Y type equivalent circuit (c) Δ type equivalent circuit.

$$\begin{cases} u_H = u_H' N_1 / N_2 \\ u_L = u_L' N_1 / N_3 \end{cases} \quad (1)$$

$$\begin{cases} i_2 = i_2' N_2 / N_1 \\ i_3 = i_3' N_3 / N_1 \end{cases} \quad (2)$$

$$\begin{cases} L_2 = L_2' (N_1 / N_2)^2 \\ L_3 = L_3' (N_1 / N_3)^2 \end{cases} \quad (3)$$

$$\begin{cases} L_{12} = L_1 + L_2 + \frac{L_1 L_2}{L_3} \\ L_{13} = L_1 + L_3 + \frac{L_1 L_3}{L_2} \\ L_{23} = L_2 + L_3 + \frac{L_2 L_3}{L_1} \end{cases} \quad (4)$$

3) The Power Flow of the Converter

According to the Δ type equivalent circuit shown in Fig. 3(c) and the existing expression of the dual-active bridge converter, the expression for the transmission power between the primary side and the HV side can be derived as (5).

$$P_{12} = \frac{4V_1 V_2}{\pi^3 f_s L_{12}} \cos(\delta_1) \cos(\delta_2) \sin(\varphi_2) \quad (5)$$

In the same way, P_{13} and P_{23} can be derived.

$$P_{13} = \frac{4V_1 V_3}{\pi^3 f_s L_{13}} \cos(\delta_1) \cos(\delta_3) \sin(\varphi_3) \quad (6)$$

$$P_{23} = \frac{4V_2 V_3}{\pi^3 f_s L_{23}} \cos(\delta_2) \cos(\delta_3) \sin(\varphi_2 - \varphi_3) \quad (7)$$

P_{12} , P_{13} , and P_{23} represent the power transmitted from primary side to HV side, primary side to LV side, and HV side to LV side, respectively. V_1 represents the input voltage, $V_1 = V_{in}$; V_2 represents the voltage of the clamping capacitor C_{Hv} on the HV side referred to the primary side, $V_2 = \frac{V_{oH} N_1}{D_H N_2}$; V_3 represents the voltage C_{Lv} on the LV side referred to the primary side, $V_3 = \frac{V_{oL} N_1}{D_L N_3}$; f_s is the switching frequency,

$$f_s = \frac{1}{T_s} \cdot \delta_1 = \frac{\pi}{2} - D_p \pi, \delta_2 = \frac{\pi}{2} - D_H \pi, \delta_3 = \frac{\pi}{2} - D_L \pi.$$

$$\begin{cases} P_2 = P_{12} - P_{23} \\ P_3 = P_{13} + P_{23} \end{cases} \quad (8)$$

P_2 and P_3 in (8) represent the power absorbed by the HV side and the LV side, respectively.

B. Related Current of the Converter

1) The transformer winding current

The GHA method is used and the current expressions are obtained as (9)-(11) [3].

$$i_{12}(t) = \frac{4}{\pi \omega L_{12}} \sum_{k=1}^{2m+1} \left[-\frac{V_1}{k^2} \cos(k\delta_1) \cos(k\omega t) + \frac{V_2}{k^2} \cos(k\delta_2) \cos(k\omega t - k\varphi_2) \right] \quad (9)$$

$$i_{13}(t) = \frac{4}{\pi \omega L_{13}} \sum_{k=1}^{2m+1} \left[-\frac{V_1}{k^2} \cos(k\delta_1) \cos(k\omega t) + \frac{V_3}{k^2} \cos(k\delta_3) \cos(k\omega t - k\varphi_3) \right] \quad (10)$$

$$i_{23}(t) = \frac{4}{\pi \omega L_{23}} \sum_{k=1}^{2m+1} \left[-\frac{V_2}{k^2} \cos(k\delta_2) \cos(k\omega t - k\varphi_2) + \frac{V_3}{k^2} \cos(k\delta_3) \cos(k\omega t - k\varphi_3) \right] \quad (11)$$

According to Fig. 3(c), the total current in each port is calculated as (12).

$$\begin{cases} i_1(t) = i_{12}(t) + i_{13}(t) \\ i_2(t) = -i_{12}(t) + i_{23}(t) \\ i_3(t) = -i_{13}(t) - i_{23}(t) \end{cases} \quad (12)$$

2) The switching currents

For the primary side, the currents of the switch devices are only related to the primary winding current of the transformer. However, for the HV/LV side, because of the existence of the output inductors, the transformer winding current is not enough to calculate the turning-on current of the switch devices. Taking S_1 and S_5 shown in Fig. 4 as examples, $i_{S1} = i_1$ and $i_{S5} = i_2 + i_{oL1}$. The output inductors current is well studied in[6]. To avoid repeating, the equation will be skipped in this paper.

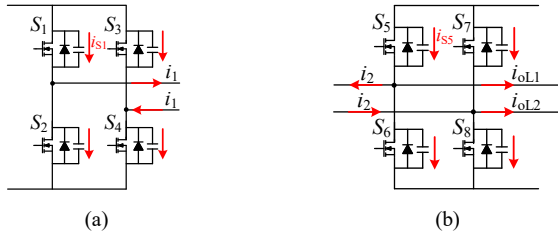


Fig. 4. ZVS current determined by transformer winding currents and output inductor currents.

C. The Optimal Control Strategy with Five Control Variables Using the Look-up Table Method

Based on the current derived in Section B, this paper proposed an optimal control strategy with five control variables using the look-up table method to minimize the switching loss and expand the range of ZVS.

To simplify the ZVS condition, the energy stored in the body capacitors of the switching devices is ignored. Based on this assumption, the implementation of ZVS depends on the direction of the current flowing through the switching device at the moment of turning-on. When the drain current i_{sx} is negative, the current flows through the anti-parallel diode at the turning-on time and this switch device S_x achieves zero-voltage turning-on and ZVS(x) in (16) ($x = 1, 2, \dots, 11, 12$, responding to 12 switch devices) is defined as 0. Conversely, the switch devices are experiencing hard turning-on and ZVS(x) is defined as 1.

In order to find the optimum control variables to minimize the switching loss under full range of output voltages and load conditions, a multivariate control scheme is adopted in this paper.

The optimization objective function related to the switching loss which account for the main part of the overall system loss under the light load conditions is listed in (16), given as the sum of the currents during the hard turning-on. Only the hard turning-on switching currents are considered because the switching loss is nearly considered to be zero when realizing ZVS.

$$F(V_{oH}, V_{oL}, D_p, D_H, D_L, \varphi_2, \varphi_3) = \sum_{x=1}^{12} |i_{sx}| \cdot \text{ZVS}(x) \quad (13)$$

The constraints in this optimization problem are given as the power flow characteristic as (17)-(18), according to (5)-(8).

$$C_1(V_{oH}, V_{oL}, D_p, D_H, D_L, \varphi_2, \varphi_3, P_{2\text{ref}}) = P_2 - P_{2\text{ref}} = 0 \quad (14)$$

$$C_2(V_{oH}, V_{oL}, D_p, D_H, D_L, \varphi_2, \varphi_3, P_{3\text{ref}}) = P_3 - P_{3\text{ref}} = 0 \quad (15)$$

To satisfy the power constraints C_1 and C_2 , two control variables are used and the remaining three are used to solve the optimization problem.

Now, the switching loss optimization problem of CF-TAB converter can be simplified to the optimization problem of multiple control variables with a single optimization target and two power constraints.

For any combination of output voltages and load conditions (V_{oH} , V_{oL} , $P_{2\text{ref}}$, $P_{3\text{ref}}$) of the CF-TAB converter, the optimization problem can be concluded as min $F(D_p, D_H, D_L, \varphi_2, \varphi_3)$ subject to 1) $C_1(D_p, D_H, D_L, \varphi_2, \varphi_3) \Big|_{V_{oH}, V_{oL}, P_{2\text{ref}}} = 0$ and 2) $C_2(D_p, D_H, D_L, \varphi_2, \varphi_3) \Big|_{V_{oH}, V_{oL}, P_{3\text{ref}}} = 0$ and 3) $0 < D_p, D_H, D_L < 1/2$, $0 < \varphi_2, \varphi_3 < \pi/2$. Then, the optimal combination of control variables (D_{p_opt} , D_{H_opt} , D_{L_opt} , φ_{2_opt} , φ_{3_opt}) that leads to the minimum optimized objective function F for a particular operating point can be obtained.

A look-up table method is used to allow the converter to operate at the minimum switching loss condition. Sweeping through the full range of output voltages and load power conditions (V_{oH} , V_{oL} , $P_{2\text{ref}}$, $P_{3\text{ref}}$), five four-dimensional (4-D) tables are generated for each of the five control variables and stored in the controller. The input parameters of the look-up table are the operating point of the converter: (V_{oH} , V_{oL} , $P_{2\text{ref}}$, $P_{3\text{ref}}$).

III. EXPERIMENTAL RESULTS AND ANALYSIS

In order to verify the advantages of the proposed control strategy applied in CF-TAB converter, a hardware experimental platform of 200-500V/3kW and 8-16V/1kW was built, and the experimental results and comparison of the DPS control and the proposed control are presented and discussed in this section.

The critical parameters of CF-TAB converter are listed in TABLE I.

The optimization effect of the proposed control strategy for the objective function (16) related to the switching loss are shown in Fig. 5. Compared with the traditional DPS control strategy, the value of the objective function is significantly reduced for a full range of voltage and load power conditions at both HV and LV side.

TABLE I. THE CRITICAL PARAMETERS OF CF-TAB CONVERTER

Circuit Parameters	Values
Input DC Voltage (V_{in})	750V
HV side Output Port Voltage (V_{oH})	200-500V
LV side Output Port Voltage (V_{oL})	8-16V
Turns ratio ($N_1: N_2: N_3$)	7.5:7.5:1
Leakage Inductances (L_1, L_2, L_3)	40μH, 40μH, 0.8μH
HV side Output Port Inductances (L_{o1}, L_{o2})	400μH
LV side Output Port Inductances (L_{o3}, L_{o4})	250μH
Switching Frequency (f_s)	100kHz

Fig. 5(a) shows variation of the objective function over full range of the HV side voltage with three different LV side voltage and power conditions while keeping the HV side load power at 300W using different control strategies. It can be seen that compared with the traditional DPS control, the proposed control algorithm helps to achieve the minimized objective function.

Furthermore, Fig. 6 illustrates the impact of the proposed switching loss minimization optimization algorithm on the

extension of ZVS range. It is evident that, despite being primarily designed to minimize the objective function related to switching losses, the proposed control strategy also extends the ZVS range of the switch transistors across a broad range of voltage and load power conditions compared with the DPS control.

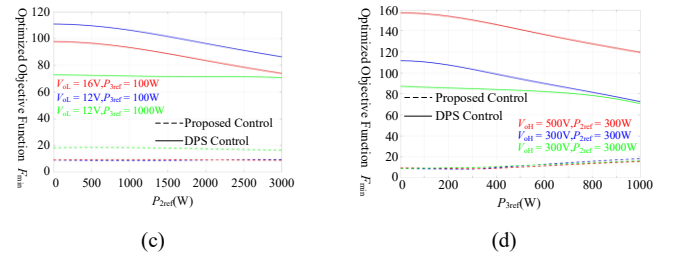
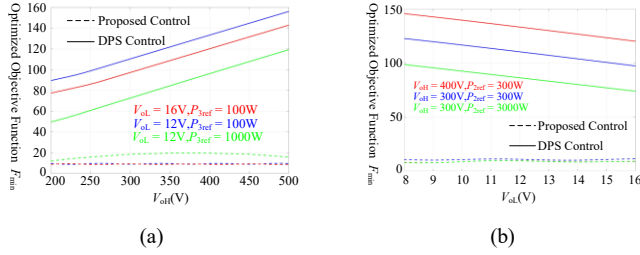


Fig. 5. Optimized objective function $F_{\min}$ using different control strategies a) $F_{\min}$ vs. V_{oH} . b) $F_{\min}$ vs. V_{oL} . c) $F_{\min}$ vs. P_2 . d) $F_{\min}$ vs. P_3 .

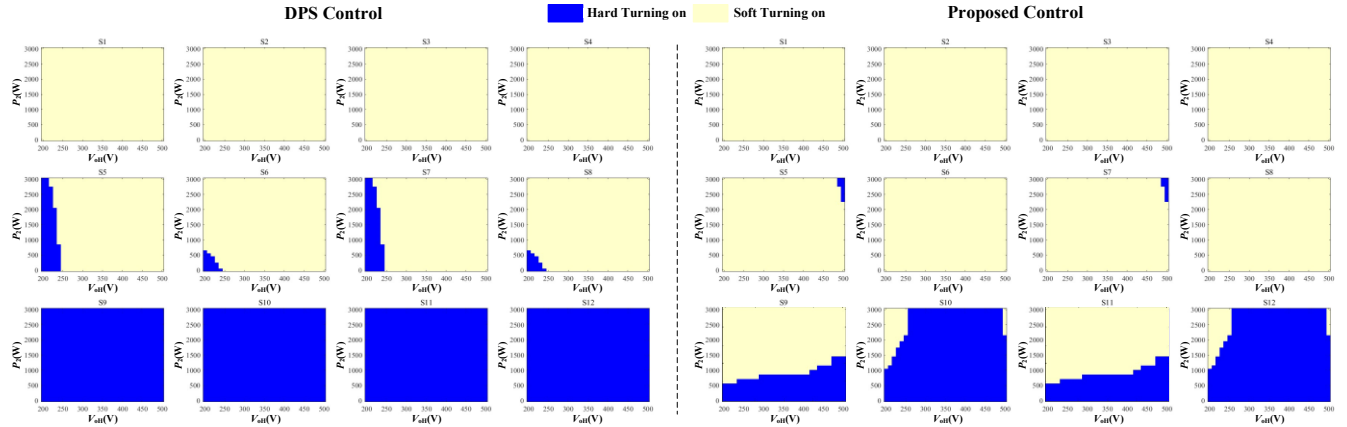


Fig. 6. Comparison of ZVS zones with DPS control and proposed control for a wide range of voltage and load power at secondary HV side while keeping the voltage $V_{oL} = 12V$ and load power $P_3 = 100W$ at secondary LV side.

The voltage and current waveforms of the transformer under DPS control and the proposed optimal control are shown in Fig. 7 and Fig. 8. The working condition of the converter ($V_{oH}, V_{oL}, P_{2ref}, P_{3ref}$) = (300V, 12V, 600W, 200W). By comparing the waveform diagrams, it can be concluded that compared to traditional DPS control, the proposed optimized control strategy achieves reductions of 10.35%, 30.76% and 20.6% in the peak currents of the three transformer ports, respectively.

Experimental waveforms of the driving voltage u_{GS} , drain-source voltage u_{DS} of the low-voltage switches S_9 - S_{12} under the operating conditions of the converter ($V_{oH}, V_{oL}, P_{2ref}, P_{3ref}$) = (300V, 12V, 400W, 120W) are shown in Fig. 9 and Fig. 10. When the traditional DPS control is adopted, the four switches S_9 - S_{12} are undergoing hard turning on; when the proposed optimal control strategy is adopted, the four switches S_9 - S_{12} are undergoing complete ZVS turning on.

To verify the advantages of the interleaved parallel structure of the converter, Fig. 11 presents the experimental waveforms of the interleaved parallel inductor current and the output current at the low-voltage port on the secondary side. Both the output current and the interleaved parallel inductor currents operate in continuous conduction mode (CCM). The interleaved parallel inductor currents exhibit a certain degree of complementarity, with a current ripple of 5 A. The output

current is the sum of the interleaved parallel inductor currents, with a ripple of 2A, corresponding to a ripple ratio of approximately 2.86%. The output current ripple is effectively suppressed, demonstrating the effectiveness of the interleaved parallel inductor structure.

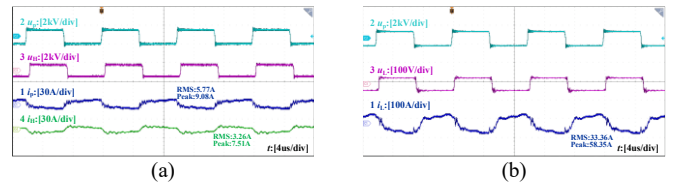


Fig. 7. Voltage and current waveforms of the transformer port under DPS control.

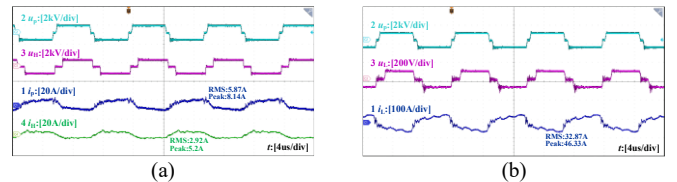


Fig. 8. Voltage and current waveforms of the transformer port under proposed control.

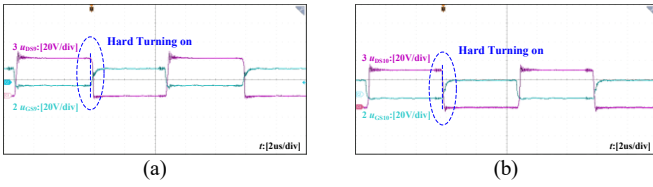


Fig. 9. Waveforms of the gate-source driving voltage (u_{GS}) and drain-source voltage (u_{DS}) under DPS control.

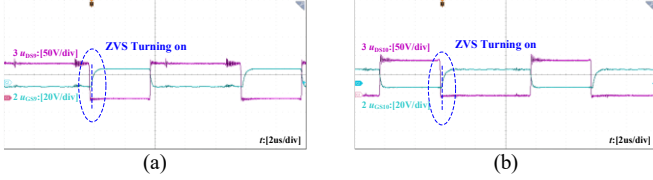


Fig. 10. Waveforms of the gate-source driving voltage (u_{GS}) and drain-source voltage (u_{DS}) under the proposed control.

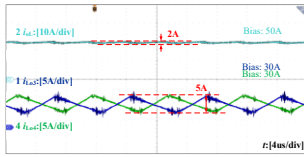


Fig. 11. Current waveforms of the interleaved parallel inductor and output current at the low-voltage port on the secondary side.

IV. CONCLUSIONS

In this paper, a current-fed triple active bridge converter for integrating the on-board charger and auxiliary power module for electric vehicles is studied. Combined with the existing derived current expressions, this paper proposed a control scheme with five control variables using the look-up table method, with the goal of reducing the optimization objective function related to the switching loss which account for the main part of the overall system loss. A 3kW/1kW hardware experimental platform was built and the effectiveness of the proposed control strategy is verified. Compared with the traditional DPS control, the control scheme proposed in this paper reduced the switching loss and realized the extension of the soft switching range of the converter.

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